

UNRESOLVED.md — novus-aexenti

Pending Issues, Optimizations & Backlog

Open Items & Tasks

1. NPU Thermal Throttling on Continuous Voice Streams:

- *Description:* When running Moonshine STT and Qwen 0.8B simultaneously during long conversations, observe if Hexagon NPU thermal throttling drops inference below 25 tok/sec.
- *Target:* Validate latency on Motorola Razr Ultra under load.

2. KV-Cache RAM Footprint on Jetson Orin Nano:

- *Description:* 9B Qwen model at 16k context window exceeds 6.2GB RAM without TurboQuant.
- *Target:* Implement 4-bit KV-cache quantization (`-ctk q4_0 -ctv q4_0`) in llama-server launch flags.

3. Mem0 Episodic Persistence vs SQLite Migration:

- *Description:* Evaluate whether in-session `rolling_habits.json` should migrate to the local SQLite `#d.u.m.b.a.s.s.` database instance for sub-millisecond retrieval.
- *Target:* Cross-link with `#d.u.m.b.a.s.s.` relational memory schema.

4. Car Wash Test Harness Validation:

- *Description:* Execute the synthetic end-to-end integration test across simulated audio ingress and execution output to measure failure recovery response.